Differentiation and Rates of Change

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Differentiation & Rates of Change Practice Exercises**

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where a and b are constants. The curve has a stationary point at $(1, 5)$. Find the coordinates of the other stationary point on the curve.

Final Answer: $(3, 1)$. To find the coordinates of the other stationary point, we first determine the unknown parameters a and b. **Step 1: Use the given coordinates of the point on the curve** Since the point $(1, 5)$ lies on the curve: $5 = a(1)^3 + b(1)^2 + 9(1) + 1$ $5 = a + b + 10 \Rightarrow a + b = -5 \quad - - -$ (Equation 1) **Step 2: Differentiate to find $\frac{dy}{dx}$** $\frac{dy}{dx} = 3ax^2 + 2bx + 9$ **Step 3: Apply the stationary condition at.

2 **[Foundational] • Target: Product and Quotient Rules**

The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where a and b are constants and $xe0$. Find the coordinates of the stationary point of the curve and determine its nature.

Final Answer: $(1, 6)$, which is a local minimum. 1. **Find the point of tangency:** Substitute $x = 2$ into the tangent line equation $y = 7x - 4$: $y = 7(2) - 4 = 10$ Since the line touches the curve at $x = 2$, the point $(2, 10)$ lies on the curve $y = ax^2 + \frac{b}{x}$: $a(2)^2 + \frac{b}{2} = 10 \Rightarrow 4a + \frac{b}{2} = 10 \Rightarrow 8a + b = 20 \quad - - -$ (1) 2. **Use the gradient of the tangent:** Differentiating $y = ax^2 + bx^{-1}$ gives: $\frac{dy}{dx} = 2ax - \frac{b}{x^2}$ The gradie.

3 **[Foundational] • Target: Chain Rule and Composite Functions**

The variables x and y are connected by the equation $y = \sqrt{2x + 5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

Final Answer: 2 units per second. We are given $y = (2x + 5)^{\frac{1}{2}}$ and $\frac{dx}{dt} = 6$. Step 1: Differentiate y with respect to x using the chain rule: $\frac{dy}{dx} = \frac{1}{2}(2x + 5)^{-\frac{1}{2}} \cdot \frac{d}{dx}(2x + 5) = \frac{1}{2\sqrt{2x+5}} \cdot 2 = \frac{1}{\sqrt{2x+5}}$ Step 2: Evaluate $\frac{dy}{dx}$ at $x = 2$.

4 **[Foundational] • Target: Tangents, Normals and Stationary Points**

The variables x and y are connected by the equation $y = 2x^3 - 5x$. Given that x increases at a constant rate of 4 units s^{-1}, find the rate of change of y at the instant when $x = 2$.

Final Answer: 76 units s^{-1}. To find the rate of change of y with respect to time t, we apply the chain rule for connected rates of change: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ First, differentiate y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$ Substitute $x = 2$ into the derivative: $\frac{dy}{dx}|_{x=2} = 6(2)^2 - 5 = 24 - 5 = 19$ Given that.

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5 [Foundational] • Target: Connected Rates of Change and Applied Optimization

A curve has equation $y = x\sqrt{16 - x^2}$ for $0 \le x \le 4$. A point $P(x, y)$ moves along the curve in the first quadrant, and the normal to the curve at P intersects the x-axis at the point N. Given that the x-coordinate of P is increasing at a constant rate of $\sqrt{3}$ units s^{-1}, find the rate of change of the area of triangle OPN, where O is the origin, at the instant when $x = 2$.

Final Answer: -6 units^2 s^{-1}. Step 1: Differentiate $y = x\sqrt{16 - x^2}$ using the product rule: $\frac{dy}{dx} = 1 \cdot \sqrt{16 - x^2} + x \cdot \frac{-2x}{2\sqrt{16 - x^2}} = \frac{16 - 2x^2}{\sqrt{16 - x^2}}$ At $x = 2$: $y = 2\sqrt{16 - 4} = 4\sqrt{3}$ $\frac{dy}{dx} = \frac{16 - 2(4)}{\sqrt{12}} = \frac{8}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$ Step 2: Find the coordinates of N, the x-intercept of the normal. The gradient of the normal at P is.

6 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2. During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm$^3 s^{-1}$, at the instant when the height of the cylindrical section is 16.5 cm.

Final Answer: 20π. 1. **Express the surface area constraint:** The total exterior surface area consists of the flat circular base, the curved cylindrical wall, and the hemispherical dome: $S = \pi r^2 + 2\pi rh + 2\pi r^2 = 3\pi r^2 + 2\pi rh$ Given $S = 180\pi$: $3\pi r^2 + 2\pi rh = 180\pi \Rightarrow 3r^2 + 2rh = 180$ 2. **Find the value of r when $h = 16.5$ cm:** $3r^2 + 2(16.5)r = 180 \Rightarrow 3r^2 + 33r - 180 = 0$ Dividing by 3: $r^2 + 11r - 60 = 0 \Rightarrow (r + 15)(r - 4) = 0$ Since.

7 [Foundational] • Target: Product and Quotient Rules

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.

Final Answer: 57 units per second. To find the rate of change of y with respect to time, apply the chain rule for connected rates of change: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ 1. Differentiate y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$ 2. Evaluate $\frac{dy}{dx}$ at $x = 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Chain Rule and Composite Functions

A curve has equation $y = \frac{kx}{2x-1}$, where $x eq \frac{1}{2}$ and k is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that x is increasing at a rate of 0.6 units per second, find the rate of change of y when $x = 2$.

Final Answer: -0.2 units per second. Step 1: Find the derivative $\frac{dy}{dx}$ in terms of k using the quotient rule. Let $u = kx$ and $v = 2x - 1$. Then $u' = k$ and $v' = 2$. $\frac{dy}{dx} = \frac{k(2x-1) - 2(kx)}{(2x-1)^2} = \frac{2kx - k - 2kx}{(2x-1)^2} = \frac{-k}{(2x-1)^2}$ Step 2: Find the parameter k. At $x = 1$, the gradient of the tangent is: $m_T = \frac{dy}{dx}\Big|_{x=1} = \frac{-k}{(2(1)-1)^2} = -k$ The gradient of the normal is given by.

9 [Intermediate] • Target: Tangents, Normals and Stationary Points

Find the equation of the tangent line to the curve $y = (4x + 1)^3$ at the point where $x = 0$.

Final Answer: $y = 12x + 1$. At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(4x + 1)^2 \cdot (4) = 12(4x + 1)^2$. At $x = 0$, slope $m = 12$. Equation: $y - 1 = 12(x - 0) \Rightarrow y = 12x + 1$.

10 [Intermediate] • Target: Connected Rates of Change and Applied Optimization

Find the equation of the tangent line to the curve $y = (5x + 1)^3$ at the point where $x = 0$.

Final Answer: $y = 15x + 1$. At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(5x + 1)^2 \cdot (5) = 15(5x + 1)^2$. At $x = 0$, slope $m = 15$. Equation: $y - 1 = 15(x - 0) \Rightarrow y = 15x + 1$.

11 [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the equation of the tangent line to the curve $y = (6x + 1)^3$ at the point where $x = 0$.

Final Answer: $y = 18x + 1$. At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(6x + 1)^2 \cdot (6) = 18(6x + 1)^2$. At $x = 0$, slope $m = 18$. Equation: $y - 1 = 18(x - 0) \Rightarrow y = 18x + 1$.

12 [Intermediate] • Target: Product and Quotient Rules

Find the equation of the tangent line to the curve $y = (7x + 1)^3$ at the point where $x = 0$.

Final Answer: $y = 21x + 1$. At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(7x + 1)^2 \cdot (7) = 21(7x + 1)^2$. At $x = 0$, slope $m = 21$. Equation: $y - 1 = 21(x - 0) \Rightarrow y = 21x + 1$.

13 [Intermediate] • Target: Chain Rule and Composite Functions

Find the equation of the tangent line to the curve $y = (8x + 1)^3$ at the point where $x = 0$.

Final Answer: $y = 24x + 1$. At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(8x + 1)^2 \cdot (8) = 24(8x + 1)^2$. At $x = 0$, slope $m = 24$. Equation: $y - 1 = 24(x - 0) \Rightarrow y = 24x + 1$.

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14 [Intermediate] • Target: Tangents, Normals and Stationary Points

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.

Final Answer: $y = 6x + 1$. At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(2x + 1)^2 \cdot (2) = 6(2x + 1)^2$. At $x = 0$, slope $m = 6$. Equation: $y - 1 = 6(x - 0) \Rightarrow y = 6x + 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 8.01 Integration and Kinematics

Find the equation of the tangent line to the curve $y = (3x + 1)^3$ at the point where $x = 0$.

Final Answer: $y = 9x + 1$. At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(3x + 1)^2 \cdot (3) = 9(3x + 1)^2$. At $x = 0$, slope $m = 9$. Equation: $y - 1 = 9(x - 0) \Rightarrow y = 9x + 1$.

16 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 8$ cm? Show your complete algebraic work.

Final Answer: 128π cm^3/s. $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 8$, $\frac{dV}{dt} = 4\pi(8)^2(0.5) = 2\pi(64) = 128\pi$ cm^3/s.

17 [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 9$ cm? Show your complete algebraic work.

Final Answer: 162π cm^3/s. $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 9$, $\frac{dV}{dt} = 4\pi(9)^2(0.5) = 2\pi(81) = 162\pi$ cm^3/s.

18 [Challenging] • Target: Chain Rule and Composite Functions • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 4$ cm? Show your complete algebraic work.

Final Answer: 32π cm^3/s. $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 4$, $\frac{dV}{dt} = 4\pi(4)^2(0.5) = 2\pi(16) = 32\pi$ cm^3/s.

19 [Challenging] • Target: Tangents, Normals and Stationary Points • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 5$ cm? Show your complete algebraic work.

Final Answer: 50π cm^3/s. $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 5$, $\frac{dV}{dt} = 4\pi(5)^2(0.5) = 2\pi(25) = 50\pi$ cm^3/s.

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20 [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 6$ cm? Show your complete algebraic work.

Final Answer: 72π cm^3/s. $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 6$, $\frac{dV}{dt} = 4\pi(6)^2(0.5) = 2\pi(36) = 72\pi$ cm^3/s.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \(x \): \(y = 3x^{3} - 5x + 7 \).

a. \(\frac{dy}{dx} = 3x^{2} - 5 \)

b. \(\frac{dy}{dx} = 9x^{2} \)

c. \(\frac{dy}{dx} = 9x^{3} - 5 \)

d. **\(\frac{dy}{dx} = 9x^{2} - 5 \)** ✓

Apply the power rule \(\frac{d}{dx}[x^n] = n x^{n-1} \): \(\frac{dy}{dx} = 3(3)x^{2} - 5 = 9x^{2} - 5 \).

22 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \(x \): \(y = 6x^{4} - 5x + 7 \).

a. **\(\frac{dy}{dx} = 24x^{3} - 5 \)** ✓

b. \(\frac{dy}{dx} = 24x^{4} - 5 \)

c. \(\frac{dy}{dx} = 6x^{3} - 5 \)

d. \(\frac{dy}{dx} = 24x^{3} \)

Apply the power rule \(\frac{d}{dx}[x^n] = n x^{n-1} \): \(\frac{dy}{dx} = 6(4)x^{3} - 5 = 24x^{3} - 5 \).

23 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \(x \): \(y = 9x^{5} - 5x + 7 \).

a. **\(\frac{dy}{dx} = 45x^{4} - 5 \)** ✓

b. \(\frac{dy}{dx} = 45x^{5} - 5 \)

c. \(\frac{dy}{dx} = 9x^{4} - 5 \)

d. \(\frac{dy}{dx} = 45x^{4} \)

Apply the power rule \(\frac{d}{dx}[x^n] = n x^{n-1} \): \(\frac{dy}{dx} = 9(5)x^{4} - 5 = 45x^{4} - 5 \).

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24 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to x: $y = 12x^{6} - 5x + 7$.

a. $\frac{dy}{dx} = 72x^{5}$

b. $\frac{dy}{dx} = 12x^{5} - 5$

c. $\frac{dy}{dx} = 72x^{6} - 5$

d. **$\frac{dy}{dx} = 72x^{5} - 5$** ✓

Apply the power rule $\frac{d}{dx}[x^n] = n x^{n-1}$: $\frac{dy}{dx} = 12(6)x^{5} - 5 = 72x^{5} - 5$.

25 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to x: $y = 15x^{7} - 5x + 7$.

a. $\frac{dy}{dx} = 105x^{6}$

b. $\frac{dy}{dx} = 15x^{6} - 5$

c. $\frac{dy}{dx} = 105x^{7} - 5$

d. **$\frac{dy}{dx} = 105x^{6} - 5$** ✓

Apply the power rule $\frac{d}{dx}[x^n] = n x^{n-1}$: $\frac{dy}{dx} = 15(7)x^{6} - 5 = 105x^{6} - 5$.

26 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to x: $y = 18x^{8} - 5x + 7$.

a. $\frac{dy}{dx} = 144x^{7}$

b. **$\frac{dy}{dx} = 144x^{7} - 5$** ✓

c. $\frac{dy}{dx} = 144x^{8} - 5$

d. $\frac{dy}{dx} = 18x^{7} - 5$

Apply the power rule $\frac{d}{dx}[x^n] = n x^{n-1}$: $\frac{dy}{dx} = 18(8)x^{7} - 5 = 144x^{7} - 5$.

27 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.

a. $y = 2x + 1$

b. $y = 6x - 1$

c. **$y = 6x + 1$** ✓

d. $y = 6x + 3$

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(2x+1)^2 \cdot (2) = 6(2x+1)^2$. At $x = 0$, slope $m = 6$. Equation: $y - 1 = 6(x - 0) \implies y = 6x + 1$.

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28 [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Find the equation of the tangent line to the curve \(y = (3x + 1)^3 \) at the point where \(x = 0 \).

a. \(y = 9x + 3 \)

b. \(y = 9x - 1 \)

c. \(y = 3x + 1 \)

d. **\(y = 9x + 1 \)** ✓

At \(x = 0 \), \(y = 1^3 = 1 \). Derivative: \(\frac{dy}{dx} = 3(3x+1)^2 \cdot (3) = 9(3x+1)^2 \). At \(x = 0 \), slope \(m = 9 \). Equation: \(y - 1 = 9(x - 0) \implies y = 9x + 1 \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Product and Quotient Rules

Differentiate $y = x^2e^{3x}$ with respect to x.

a. $xe^{3x}(3x + 2)$ ✓

b. $2xe^{3x}$

c. $3x^2e^{3x}$

d. $x^2e^{3x} + 2x$

Product rule: $y' = (2x)(e^{3x}) + (x^2)(3e^{3x}) = xe^{3x}(2 + 3x)$.

30 [Intermediate] • Target: Product and Quotient Rules

Find $\frac{dy}{dx}$ for $y = \frac{2x-1}{x+4}$.

a. $\frac{9}{(x+4)^2}$ ✓

b. $\frac{2}{(x+4)^2}$

c. $\frac{7}{(x+4)^2}$

d. $\frac{2x+9}{(x+4)^2}$

Quotient rule: $\frac{2(x+4)-(2x-1)(1)}{(x+4)^2} = \frac{2x+8-2x+1}{(x+4)^2} = \frac{9}{(x+4)^2}$.

31 [Intermediate] • Target: Product and Quotient Rules

Find the derivative of $f(x) = x\ln x$ evaluated at $x = e$.

a. 2 ✓

b. 1

c. e

d. 0

$f'(x) = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$. At $x = e$: $f'(e) = \ln e + 1 = 1 + 1 = 2$.

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32 [Intermediate] • Target: Chain Rule and Composite Functions

Differentiate $y = (3x^2 - 5)^4$ with respect to x.

a. $24x(3x^2 - 5)^3$ ✓

b. $12x(3x^2 - 5)^3$

c. $4(3x^2 - 5)^3$

d. $18x^2(3x^2 - 5)^3$

Chain rule: $\frac{dy}{dx} = 4(3x^2 - 5)^3 \cdot (6x) = 24x(3x^2 - 5)^3$.

33 [Intermediate] • Target: Chain Rule and Composite Functions

Find the derivative of $y = \sin^3(2x)$.

a. $6\sin^2(2x)\cos(2x)$ ✓

b. $3\sin^2(2x)\cos(2x)$

c. $6\sin(2x)\cos(2x)$

d. $-6\sin^2(2x)\cos(2x)$

Chain rule: $3\sin^2(2x) \cdot \cos(2x) \cdot 2 = 6\sin^2(2x)\cos(2x)$.

34 [Intermediate] • Target: Chain Rule and Composite Functions

Find the gradient of $y = \sqrt{4x + 9}$ at the point where $x = 4$.

a. $\frac{2}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. 5

$\frac{dy}{dx} = \frac{1}{2}(4x + 9)^{-1/2}(4) = \frac{2}{\sqrt{4x+9}}$. At $x = 4$: $\frac{2}{\sqrt{25}} = \frac{2}{5}$.

35 [Intermediate] • Target: Tangents, Normals and Stationary Points

Find the equation of the tangent to the curve $y = x^2 - 4x + 3$ at the point where $x = 3$.

a. $y = 2x - 6$ ✓

b. $y = 2x + 6$

c. $y = -2x$

d. $y = x - 3$

At $x = 3$: $y = 9 - 12 + 3 = 0$. $\frac{dy}{dx} = 2x - 4$. At $x = 3$, $m = 2$. Line: $y - 0 = 2(x - 3) \Rightarrow y = 2x - 6$.

36 [Intermediate] • Target: Tangents, Normals and Stationary Points

Find the coordinates of the stationary point of $y = x^3 - 3x + 2$ in the first quadrant.

a. $(1, 0)$ ✓

b. $(1, 2)$

c. $(2, 4)$

d. $(0, 2)$

$\frac{dy}{dx} = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = 1$ (quadrant 1). $y = 1 - 3 + 2 = 0$. Point is $(1, 0)$.

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37 [Intermediate] • Target: Tangents, Normals and Stationary Points

Determine the nature of the stationary point of $f(x) = 2x^3 - 9x^2 + 12x$ at $x = 2$.

a. **Local Minimum** ✓

b. Local Maximum

c. Point of Inflexion

d. Vertical Asymptote

$f''(x) = 12x - 18$. At $x = 2$: $f''(2) = 24 - 18 = 6 > 0$, confirming a local minimum.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The radius of a circular oil spill is increasing at 0.5 m/s. At what rate is the area increasing when the radius is 10 m?

a. 10π m^2/s ✓

b. 20π m^2/s

c. 5π m^2/s

d. 100π m^2/s

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(10)(0.5) = 10\pi$ m^2/s.

39 [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A solid cylindrical metal rod of radius r and length h has fixed volume $V = 54\pi$. Express the surface area S in terms of r and find r minimizing S.

a. $r = 3$ ✓

b. $r = 2$

c. $r = 6$

d. $r = 4$

$V = \pi r^2 h = 54\pi \Rightarrow h = 54/r^2$. $S = 2\pi r^2 + 2\pi r(54/r^2) = 2\pi r^2 + 108\pi/r$.

$\frac{dS}{dr} = 4\pi r - 108\pi/r^2 = 0 \Rightarrow 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

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40 [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

If $y = 3x^2 - 4x$ and x increases at a rate of 2 units/s, find the rate of increase of y when $x = 5$.

a. 52 units/s ✓

b. 26 units/s

c. 30 units/s

d. 60 units/s

$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} = (6x - 4)(2)$. At $x = 5$: $(26)(2) = 52$.

41 [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 7$ cm?

a. 98π cm^3/s ✓

b. 196π cm^3/s

c. $\frac{98}{2}\pi$ cm^3/s

d. 49π cm^3/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 7$, $\frac{dV}{dt} = 4\pi(7)^2(0.5) = 2\pi(49) = 98\pi$ cm^3/s.

42 [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 8.01 Integration and Kinematics

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 3$ cm?

a. 18π cm^3/s ✓

b. 36π cm^3/s

c. 9π cm^3/s

d. $\frac{18}{2}\pi$ cm^3/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 3$, $\frac{dV}{dt} = 4\pi(3)^2(0.5) = 2\pi(9) = 18\pi$ cm^3/s.

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43 [Challenging] • Target: Chain Rule and Composite Functions • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the x-coordinates of the stationary points of $y = x^3 + 15/2x^2 + 18x$.

a. $x = -3$ or $x = 2$

b. $x = -3$ or $x = -2$ ✓

c. $x = -2$ or $x = -2$

d. $x = 3$ or $x = -2$

$\frac{dy}{dx} = 3x^2 + 15x + 18 = 3(1)(x + 2)(x + 3) = 0 \Rightarrow x = -3$ or $x = -2$

44 [Challenging] • Target: Tangents, Normals and Stationary Points • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the x-coordinates of the stationary points of $y = 2x^3 + 12x^2 + 18x$.

a. $x = -3$ or $x = -1$ ✓

b. $x = -2$ or $x = -1$

c. $x = 3$ or $x = -1$

d. $x = -3$ or $x = 1$

$\frac{dy}{dx} = 6x^2 + 24x + 18 = 3(2)(x + 3)(x + 1) = 0 \Rightarrow x = -3$ or $x = -1$

45 [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the x-coordinates of the stationary points of $y = 2x^3 - 9x^2 - 24x$.

a. $x = 0$ or $x = 4$

b. $x = 1$ or $x = 4$

c. $x = -1$ or $x = -4$

d. $x = -1$ or $x = 4$ ✓

$\frac{dy}{dx} = 6x^2 - 18x - 24 = 3(2)(x - 4)(x + 1) = 0 \Rightarrow x = -1$ or $x = 4$

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